

L1. Information, Entropy, and Computational Complexity

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Outline

- How to Quantify Information?
- Computational Complexity
 - Decision Problems: P vs NP
 - Randomness
 - Quantum Randomness
- A Few Milestones

1.1 How to Quantify Information?

Take-Home Messages

- Shannon quantified the information content of a message by entropy, which comes from a physical quantity that describes order/disorder.
- The erasure of information is irreversible, which must dissipates heat into environment.
- As an example, Shannon entropy provides a lower bound for the coding efficiency in data compression.

Shannon Entropy

- Shannon (1948) developed the theory of information in the context of sending information over a channel such as a telephone wire.
 - The amount of information in an event x must depend only upon its probability p_x .
 - Continuity: $I(p)$ is a continuous function of p .
 - Additivity: $I(p_x, p_y) = I(p_x) + I(p_y)$ for independent events x and y .
- Up to an additive and multiplicative constant (i.e. an affine transformation), Shannon proved that there is a unique measure, known as the **Shannon entropy**, which satisfied these requirements.

- Due to additivity, $I(p_x, p_y) = I(p_x) + I(p_y)$ for independent events x and y .
 - In particular, if $y = x$, we have $I(p_x, p_x) = 2I(p_x)$.
 - The joint probability of events x and y is $p = p_x p_y$.
 - Consequently, $I(p_x^2) = 2I(p_x)$.
- In general, for any integer s , additivity implies

$$I(p_x^s) = I(p_x^{s-1}, p_x) = I(p_x^{s-1}) + I(p_x) = \cdots = sI(p_x).$$
 - With $p = p_x^t$ for integer t , we have $I(p^{1/t}) = (1/t)I(p)$.
- Consequently, for any rational $r = s/t$,

$$I(p_x^r) = I(p_x^{s/t}) = sI(p_x^{1/t}) = s(1/t)I(p_x) = rI(p_x).$$

- Therefore, continuity yields

$$I(p_x) = I\left((1/2)^{-\log_2(p_x)}\right) = -\log_2(p_x)I(1/2).$$

- $I(1/2)$ is a multiplicative constant, which can be taken as the unit of the information measure.

Shannon Entropy

If X is a random variable over a set of events x such that event x occurs with probability p_x , then the Shannon entropy of event x is $-\log_2(p_x)$.

Information is Physical

- [Claude Shannon] If we consider a set of possible events whose probabilities of occurrence are $p_1, p_2, \dots, p_n$, the average information is

$$H = \langle I(p_i) \rangle \sim - \sum_{i=1}^n p_i \log_2 p_i.$$

- Consider a register which stores 1 bit of information and is connected to a thermal reservoir of temperature T . The bit can be either 1 or 0. So we have $p_0 = 1/2$, $p_1 = 1/2$, so $H = - \sum_i p_i \log_2 p_i = 1$.

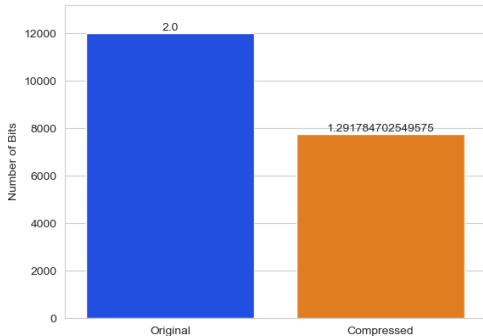
- We can erase it by setting the bit to zero. After erasure, $p_0 = 1$, $p_1 = 0$, so $H = -\sum_i p_i \log_2 p_i = 0$.
- Erasure is irreversible! To erase, we must dissipate heat in the surroundings. The entropy of the surroundings must go up by at least $k_B \ln 2$. This result, which demonstrates the intimate link between thermodynamics and information theory, has been verified experimentally [Berut et al., Nature 483, 187 (2012)].
- [John von Neumann] *"It is already in use under that name. . . . and besides, it will give you great edge in debates because nobody really knows what entropy is anyway."*

Example: Data Compression

- As an application of the Shannon entropy, we consider classical information stored in the form of binary digits 0 (with probability $p = 0.9$) and 1 (with probability $1 - p$).
- We can develop a two-bit encoding scheme:

Encoding	Length	Probability
00 $\rightarrow$ 0	1	$p^2 = 0.81$
01 $\rightarrow$ 10	2	$p(1 - p) = 0.09$
10 $\rightarrow$ 110	3	$p(1 - p) = 0.09$
11 $\rightarrow$ 1110	4	$(1 - p)^2 = 0.01$

Lab Example: Data Compression



- The average length reduces from 2 to

$$\bar{L} = 0.81 + 2 \times 0.09 + 3 \times 0.09 + 4 \times 0.01 = 1.3.$$

- In general, we model a set of data to be compressed by a random variable X . Is there a lower bound above which we can compress, at least in principle?
- In the sequence $X_1, X_2, \dots, X_n$, the expected number of occurrences of $X = x$ is $nP(X = x)$. Hence,

$$P(X_1, X_2, \dots, X_n) = \prod_i P(X_i) \approx \prod_x P(X = x)^{nP(X=x)}.$$

- The information content is approximately

$$-\log_2 P(X_1, X_2, \dots, X_n) = -n \sum_x P(X = x) \log_2 P(X = x).$$

- As the Shannon entropy of X is defined as the expected logarithm of the probability:

$$H(X) = - \sum_x P(X = x) \log_2 P(X = x),$$

We obtain

$$P(X_1, X_2, \dots, X_n) \approx 2^{-nH(X)}.$$

- This shows that there are at most 2^{nH} typical sequences, and, hence, it only requires $nH(X)$ bits to encode them.

Information and Physics

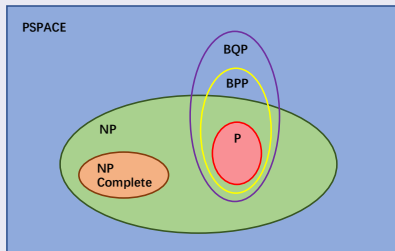
- The development of computer science led to a new way of thinking about physics: physical systems can be thought of as computers processing information.
 - The initial state of a physical system is the input to a computer.
 - The evolution of the physical system is the computation performed by the computer.
 - The final state of the system is the output of the computer.
- Classical information theory assumes that information evolves according to the law of classical (Newtonian) physics.

- In quantum information theory, on the other hand, information is represented by quantum states and allowed to evolve according to the laws of quantum mechanics.
- Note that how much information we can store and how quickly we can process it depends (only) upon the laws of physics (and not some arbitrary axioms of mathematics that come out of nowhere).
- Why do we care for a quantum computer? What do we mean by it being faster than a classical computer?

1.2 Computational Complexity

Take-Home Messages

Problems exist that are thought to be intractable in the classical regime but tractable in bounded-error polynomial time for a quantum computer.



How Fast Can We Compute?

- For example, sorting is one of the most common problem we encounter. A simple algorithm is known as compare-and-swap.
- The pseudo code is something like this:

```
for j = 1 to n-1
  for k = j+1 to n
    compare-and-swap(list[j],list[k])
  end k
end j
```


- This algorithm sorts a list of n entries into the correct order. The number of compare-and-swap operations executed by the algorithm is

$$(n - 1) + (n - 2) + \cdots + 1 = n(n - 1)/2.$$

- In other words, the algorithm uses $O(n^2)$ compare-and-swap operations.¹
- Can we do better? Algorithms such as ‘heapsort’ uses $O(n \log n)$ compare-and-swap operations.

¹An upper bound up to an unimportant constant factor.

- Computational complexity is the study of time and space resources required to solve computational problems.
- A problem is regarded as *easy* or *tractable* if an algorithm using polynomial resources exist, and as *hard* or *intractable* if the best algorithm requires exponential resources.

complexity	$n = 10$	$n = 20$	$n = 30$	$n = 40$	$n = 50$
n	0.00001 s	0.00002 s	0.00003 s	0.00004 s	0.00005 s
n^3	0.001 s	0.008 s	0.027 s	0.064 s	0.125 s
n^5	0.1 s	3.2 s	24.3 s	1.7 m	5.2 m
2^n	0.001 s	1.0 s	17.9 m	12.7 d	35.7 y
3^n	0.059 s	58 m	6.5 y	3.9×10^5 y	2×10^{10} y

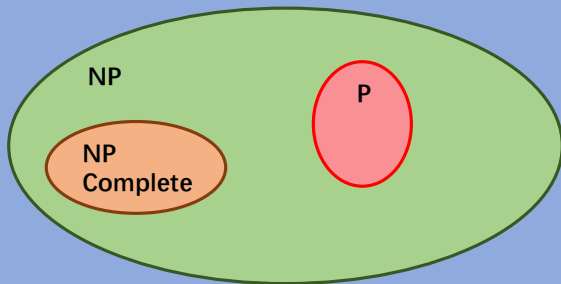
Decision Problems: P vs NP

- Most computational problems can be formulated as decision problems, whose answer is yes or no. For example, the *factoring decision problem* is: Given integer m and $l < m$, does m have a non-trivial factor less than l ?
- The class of all problems computable in polynomial time $\text{poly}(n)$ (on a deterministic machine) is denoted by **P**. This means we have an effective algorithm to solve the problem.
- The factoring decision problem is believed not to be in **P**, meaning that it is hard to solve. But, interestingly, if someone claims f is a factor (known as a *witness*), checking it (by, e.g., long-division) is easy.

- The class **NP** is defined as the class of all decision problems that can be solved by polynomial time nondeterministic algorithms.
 - A nondeterministic algorithm that solves a decision problem is said to operate in polynomial time if there exists a polynomial p such that for every instance there is some guess that can be verified within time $p(n)$.
 - The class NP does not imply the existence of a polynomial time method for solving decision problems. Hence, potentially hard problems.
 - Another example is the traveling salesman problem: Given a set of n cities, the distances between them, and a bound B , does there exist a tour of all the cities having total length B or less?

- By definition, every decision problem solvable by a polynomial time deterministic algorithm is also solvable by a polynomial time nondeterministic algorithm.
- Our tentative understanding of the computational complexity theory is $P \subseteq NP$.
 - For many individual problems in NP, such as the traveling salesman problem, no polynomial time solution algorithms have been found.
 - However, we have no proof that $P \neq NP$, or $P = NP$.
 - The most difficult problems in NP are known as NP-complete, the traveling salesman problem being one of them. [See a list in, e.g., M. R. Garey and D. S. Johnson, *Computers and intractability—A guide to the theory of NP-completeness* (Freeman, New York, 1979).]

PSPACE

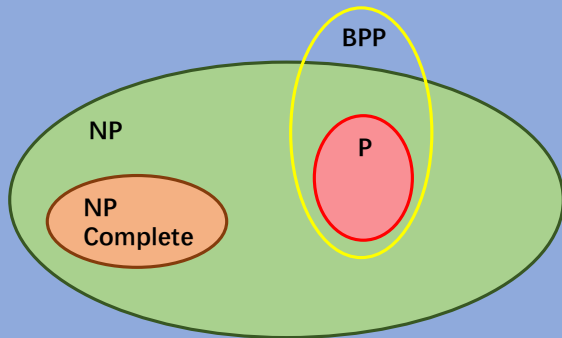


Randomness

- A probabilistic algorithm is somewhat similar to a nondeterministic one. The difference is that choice is produced by randomness, like coin tossing, not by guessing.
- Informally, a problem is in **BPP** (bounded-error probabilistic polynomial time) if there is an algorithm for it that has the following properties:
 - It is allowed to flip coins and make random decisions.
 - It is guaranteed to run in polynomial time.
 - On any given run of the algorithm, it has a probability of at most $1/3$ of giving the wrong answer, whether the answer is YES or NO.

- Obviously, $P \subseteq BPP$. There are problems known to be in BPP but not known to be in P (i.e., randomness helps). However, the number of such problems is decreasing, and it is conjectured that $P = BPP$ (i.e., derandomization).
 - One of the most famous problems that was known, for a long time, to be in BPP but not known to be in P was the problem of determining whether a given number is prime. However, Manindra Agrawal and his students Neeraj Kayal and Nitin Saxena found a deterministic polynomial-time algorithm for this problem [*PRIMES is in P*, Ann. Math. 160, 781 (2004)], thus showing that it is in P .

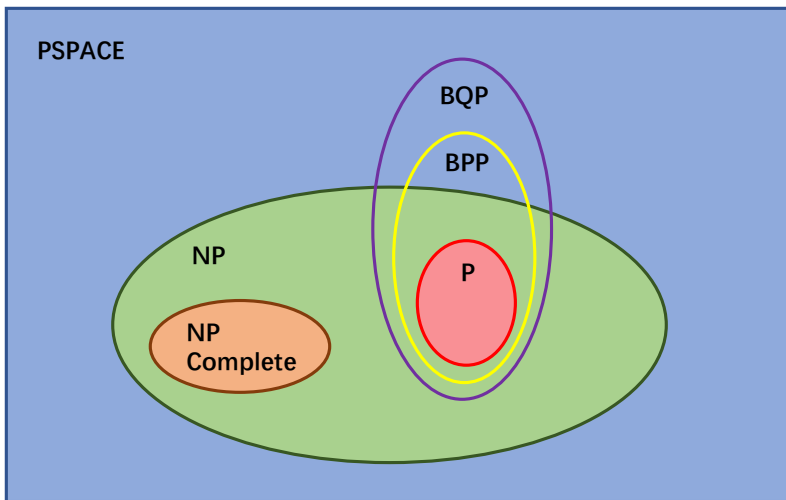
PSPACE



Quantum Randomness

- A quantum information processor differs from a classical one (a Turing machine) in two important ways.
 - The input can be prepared not only in the binary form, but also in any superposition state.
 - Intervention (such as examine the state of the machine) could modify the state of the quantum machine, and it would not be possible to resume the processing operation.
- In many ways, quantum mechanics generalizes real probability to complex probability amplitudes. One can generalize the classical BPP class to **BQP**, and $\text{BPP} \subseteq \text{BQP}$. But do we have $\text{BPP} \neq \text{BQP}$?

- A decisive case that quantum algorithms can do better than their classical counterparts is factoring, i.e., given an n -bit integer, find the prime factorization. In 1994 Peter Shor discovered a quantum algorithm that solves this in $O(n^3)$, faster than the known classical algorithm $[2^{O(n^{1/3})}]$.
- A comprehensive catalog of quantum algorithms has been maintained at <https://quantumalgorithmzoo.org> (or, <https://www.qtumist.com/quantum-algorithm-zoo> in Chinese).



1.3 A Few Milestones

- [1982] Richard Feynman's paper has become a landmark in quantum computation and simulation, and has been credited for the birth of these fields.
- [1994] Peter Shor's algorithm for factoring integers ignited a tremendous increase of interest in quantum computing.
- [2018] Google declared quantum supremacy, the potential ability of quantum computing devices to solve problems that classical computers practically cannot.
- [2019] IBM launched first 20-qubit commercial quantum computer named Q System One. Organizations started to prepare for the second quantum revolution by education in quantum engineering.

40 Years of Quantum Computing

[Nature Review Physics (2022)] ... celebrate four decades of quantum computing by looking back at the milestones of the field and forward at the challenges and opportunities that lie ahead ... put together a collection of relevant reviews, opinion and retrospective pieces published in Nature Reviews Physics.

Ref: <https://www.nature.com/collections/djhfiabiig>

Summary

- Shannon showed that entropy can uniquely (up to an affine transformation) measure the amount of information.
- Theory of classical information and computation relies on laws of classical physics, while quantum physics opens new physical possibilities for communication and information processing.
- Quantum computers can enhance our computational power, at least in problems for which we can find effective algorithms. With recent hardware development, a number of problems which are currently intractable will become efficiently solvable.

References

- S. Aaronson, Quantum Computing Since Democritus, Cambridge University Press (2013).
- J. D. Hidary, Quantum Computing: An Applied Approach, 2nd Ed. (Springer, 2021), Chap. 4.